SEPARATION PROPERTIES OF AN ALGEBRAIC SPACE IMPLY THAT ANY MORPHISM FROM IT IS SEPARATED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement has two enumerations, the second continuing the numbering at (4); both enumerations are inside the counted statement span.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Stability of the separation axioms under composition. Lemma lemma-compose-after-separated, used at proof lines 674-677. Gives parts (4), (5) and (6) by composing f with the structure morphism of X.
- Separatedness of a morphism versus of an algebraic space over a base. Spaces, Definition spaces-definition-separated and Properties of Spaces, Definition spaces-properties-definition-separated, used at proof lines 674-681. Identifies the two forms of the three properties.
- The base Spec(Z). used at proof lines 678-681. Reduces the absolute assertions to the relative ones.

Source reference. Original lemma 03KR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4687–4687. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 02X5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3828–3828. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 03BS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4359–4359. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let X be an algebraic space over S.*

(1) *If X is separated then X is separated over S.*
(2) *If X is locally separated then X is locally separated over S.*
(3) *If X is quasi-separated then X is quasi-separated over S.*

Let $f : X \to Y$ be a morphism of algebraic spaces over S.

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(4) *If X is separated over S then f is separated.*
(5) *If X is locally separated over S then f is locally separated.*
(6) *If X is quasi-separated over S then f is quasi-separated.*

Proof. Parts (4), (5), and (6) follow immediately from Lemma 03KR and Spaces, Definition 02X5. Parts (1), (2), and (3) follow from parts (4), (5), and (6) by thinking of X and Y as algebraic spaces over $\mathrm{Spec}(\mathbf{Z})$, see Properties of Spaces, Definition 03BS. □